

 VIT-AP UNIVERSITY	Final Assessment Test – Short Summer-I (2024-25) - June 2025	
	Maximum Marks: 100	Duration: 3 Hours
Course Code: CSE2008	Course Title: Operating Systems	
Set No: 2	Exam Type : Closed Book	School: SCOPE
Date: 17/06/2025	Slot: C	Session: FN
Keeping mobile phone/smart watch, even in 'off' position is treated as exam malpractice General Instructions if any: 1. "fx series" - non Programmable calculator are permitted : YES 2. Reference tables permitted : NO		

PART – A: Answer any TEN Questions, Each Question Carries 10 Marks (10×10=100 Marks)

1. What are the major activities of an Operating system with regard to process management and storage management? **(10 M)**
2. Describe the layered approach and microkernel structure in operating system design? **(10 M)**
3. Consider the following set of processes, with the length of the CPU burst given in milliseconds:

Process	Burst time	Priority
P1	3	2
P2	2	1
P3	4	4
P4	6	3

The processes are assumed to have arrived in the order P1, P2, P3, P4 all at time 0. **(10 M)**

- a) Draw four Gantt charts that illustrate the execution of these processes using the following scheduling algorithms: FCFS, SJF, non-preemptive priority (a larger priority number implies a higher priority), and RR (quantum = 4).
 - b) What is the turnaround time of each process for each of the scheduling algorithms in part a?
 - c) What is the waiting time of each process for each of these scheduling algorithms?
 - d) Which of the algorithms results in the minimum average waiting time (over all processes)?
4. Describe the differences between short-term, medium-term and long-term scheduling **(10 M)**
 5. What are the necessary conditions that hold simultaneously in the system for a deadlock situation to arise in a system? **(10 M)**
 6. Consider the snapshot of the resource allocation state of the system and solve the following questions using bankers algorithm **(10 M)**
 - i) How many resources of type A, B, C, D are there?
 - ii) What are the contents of need matrix?
 - iii) Find if the system is in safe state? If it is, find the safe sequence. If the system is unsafe show how deadlock can occur

	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	0	0	1	2	0	0	1	2	1	5	2	0
P1	1	0	0	0	1	7	5	0				
P2	1	3	5	4	2	3	5	6				
P3	0	6	3	2	0	6	5	2				
P4	0	0	1	4	0	6	5	6				

7. Consider a swapping system in which memory consists of the following whole sizes in memory order: 12 KB, 4 KB, 24 KB, 15 KB, 9 KB, 7 KB, 10 KB, and 11 KB. Which hole is taken for successive segment requests of : (i) 14 KB, (ii) 8 KB (iii) 5 KB for first fit, best fit, worst fit approaches? Rank the algorithms in terms of how efficiently they use the memory. **(10 M)**
8. Consider the following segment table:

<u>Segment</u>	<u>Base</u>	<u>Length</u>
0	219	600
1	2300	14
2	90	100
3	1327	580
4	1952	96

What are the physical addresses for the following logical addresses? **(10 M)**
a. 0,430 b. 1,10 c. 2,500 d. 3,400 e. 4,112

9. Consider the following page reference string: 6, 1, 1, 2, 0, 3, 4, 6, 0, 2, 1, 2, 1, 2, 0, 3, 2, 1, 4, 0. How many page faults would occur for the following replacement algorithms, assuming three frames? Remember that all frames are initially empty, so your first unique pages will cost one fault each.
- LRU replacement
 - FIFO replacement
 - Optimal replacement
- (10 M)**
10. Consider a disk with 170 tracks and the queue has random requests from different processes in the order: (70, 140, 50, 125, 30, 25, 160). Initially arm is at 60. Find the Average Seek length using FIFO, SSTF, SCAN and C-SCAN algorithm. **(10 M)**
11. Discuss Tree structured and Acyclic Graph file directory structures in operating system with advantages and disadvantages in each structure? **(10 M)**
12. Discuss the goals and principles of protection in a modern computer systems? **(10 M)**

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Course Code: CSE2008	Course Title: Operating Systems	
Set No: 3	Exam Type : Closed Book	School: SCOPE
Date: 14/06/2025	Slot: A	Session: FN
Keeping mobile phone/smart watch, even in 'off' position is treated as exam malpractice		
General Instructions if any: 1. "fx series" - non Programmable calculator are permitted : YES 2. Reference tables permitted : NO		

PART – A: Answer any TEN Questions, Each Question Carries 10 Marks (10×10=100 Marks)

1. What are the major activities of an Operating System regarding memory management and file management? **(10 M)**
2. Define a Process? Explain Process states and PCB(Process Control Block) in detail? Describe the actions taken by a kernel to context switch between processes? **(10 M)**
3. The following processes are being scheduled using a preemptive, round robin scheduling algorithm. Each process is assigned a numerical priority with a higher number indicating a higher relative priority. In addition to the processes listed below, the system also has an idle task (which consumes no CPU resources and is identified as Pidle). This task has priority 0 and is scheduled whenever the system has no other available processes to run. The length of a time quantum is 15 units. If a process is preempted by a higher priority process, the preempted process is placed at the end of the queue. **(10M)**

Process	Priority	Burst	Arrival
P1	40	20	0
P2	30	25	25
P3	30	25	30
P4	35	15	60
P5	5	10	100
P6	10	10	105

- A) Show the scheduling order of the process using a gantt chart
 - B) What is the turnaround time for each process
 - C) What is the waiting time of each process
4. How cooperating processes can communicate in a shared-memory environment? Explain it with a solution to producer- consumer problem having bounded buffer area? **(10 M)**
 5. What are the necessary conditions that hold simultaneously in the system for a deadlock situation to arise in a system? **(10 M)**
 6. Consider the following snapshot of a system **(10 M)**
 Answer the following questions using the banker's algorithm:
 - a. What is the content of the matrix Need?
 - b. Is the system in a safe state?
 - c. If a request from process P1 arrives for (0,4,2,0), can the request be granted immediately?

	<u>Allocation</u>	<u>Max</u>	<u>Available</u>
	<u>A B C D</u>	<u>A B C D</u>	<u>A B C D</u>
P_0	0 0 1 2	0 0 1 2	1 5 2 0
P_1	1 0 0 0	1 7 5 0	
P_2	1 3 5 4	2 3 5 6	
P_3	0 6 3 2	0 6 5 2	
P_4	0 0 1 4	0 6 5 6	

7. Given six memory partitions of 300 KB, 600 KB, 350 KB, 200 KB, 750 KB and 125 KB (in-order) how would the first-fit, best-fit and worst-fit algorithms place processes of size 115 KB, 500 KB, 358 KB, 200 KB, and 375 KB (in-order)? Rank the algorithms in-terms of how efficiently they use the memory? (10 M)
8. Consider a program consists of five segments: $S_0 = 600$, $S_1 = 14$ KB, $S_2 = 100$ KB, $S_3 = 580$ KB, and $S_4 = 96$ KB. Assume at that time, the available free space partitions of memory are 1200–1805, 50 – 150, 220-234, and 2500-3180. (10 M)
Find the following:
1. Draw logical to physical maps and segment table?
 2. Allocate space for each segment in memory?
 3. Calculate the external fragmentation and the internal fragmentation?
9. Consider the following page reference string:
7, 2, 3, 1, 2, 5, 3, 4, 6, 7, 7, 1, 0, 5, 4, 6, 2, 3, 0, 1.
Assuming demand paging with three frames, how many page faults would occur for the following replacement algorithms?
• LRU replacement • FIFO replacement • Optimal replacement (10 M)
10. A disk drive has 200 cylinders, numbered 0 to 199. The drive is currently serving a request at cylinder 60. The queue of pending requests, in FIFO order, is 87, 160, 40, 140, 36, 72, 66, 15. Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for each of the following disk-scheduling algorithms?
i) FCFS ii) SSTF iii) SCAN iv) LOOK v) C-SCAN vi) C-LOOK (10 M)
11. Discuss sequential and direct access methods in retrieving information from a file and read into computer memory? (10 M)
12. Explain how protection domains, combined with an access matrix are used to specify the resources a process may access? (10 M)